VLSI Devices
Lecture 12

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L12

How can we determine ϕ_s?

- For given V_{gs} and V, we can solve the MOS equation.

$$V_{gs} = V_{fb} + \phi_s - \frac{Q_s}{C_{ox}}$$

$$= V_{fb} + \phi_s + \frac{\sqrt{2\epsilon_{si} k_B T N_a}}{C_{ox}} \left[\frac{q\phi_s}{k_B T} + \frac{n_i^2}{N_a^2} \exp\left(\frac{q}{k_B T} (\phi_s - V) \right) \right]^{1/2}$$

Taur, Eq. (5.14)

Only two important terms are kept.

  - We can numerically solve the above equation to obtain ϕ_s. (Newton method)

Write a code. (1)

- Calculate I_d at any V_{gs} and V_{ds}.
  - We follow the Riemann sum approach.
  - Then, let's introduce 101 V values, from 0 V to V_{ds}.

$$I_d \approx q\mu_{eff}\frac{W}{L}\sum_{j=0}^{100}\left[\int_{\delta}^{\phi_s}\frac{\left(n_i^2/N_a\right)\exp\left(\frac{q}{k_BT}\left(\phi - V_j\right)\right)}{E_x\left(\phi, V_j\right)}d\phi\right]\Delta V_j$$

  - Use $q\mu_{eff}\frac{W}{L}$ = 4.8 X 10^{-17} A cm^2 V^{-1}.

Write a code. (2)

- Now, we must perform the ϕ-integral.

$$\int_{\delta}^{\phi_s} \frac{\left(n_i^2/N_a\right)\exp\left(\frac{q}{k_B T}\left(\phi - V_j\right)\right)}{E_x\left(\phi, V_j\right)} d\phi$$

  - We must know two bounds, δ and ϕ_s.
  - We can set a small value for δ. (It doesn't matter. $\delta = V_j$ + 0.1 mV, for example)
  - However, ϕ_s is not easy to evaluate. (In general, we must adopt the Newton method.)
  - Instead, an approximate value for ϕ_s will be used.

Write a code. (3)

- The ϕ-integral is approximated as

$$\int_{\delta}^{\phi_s} \frac{\left(n_i^2/N_a\right)\exp\left(\frac{q}{k_B T}\left(\phi - V_j\right)\right)}{E_x\left(\phi, V_j\right)} d\phi$$

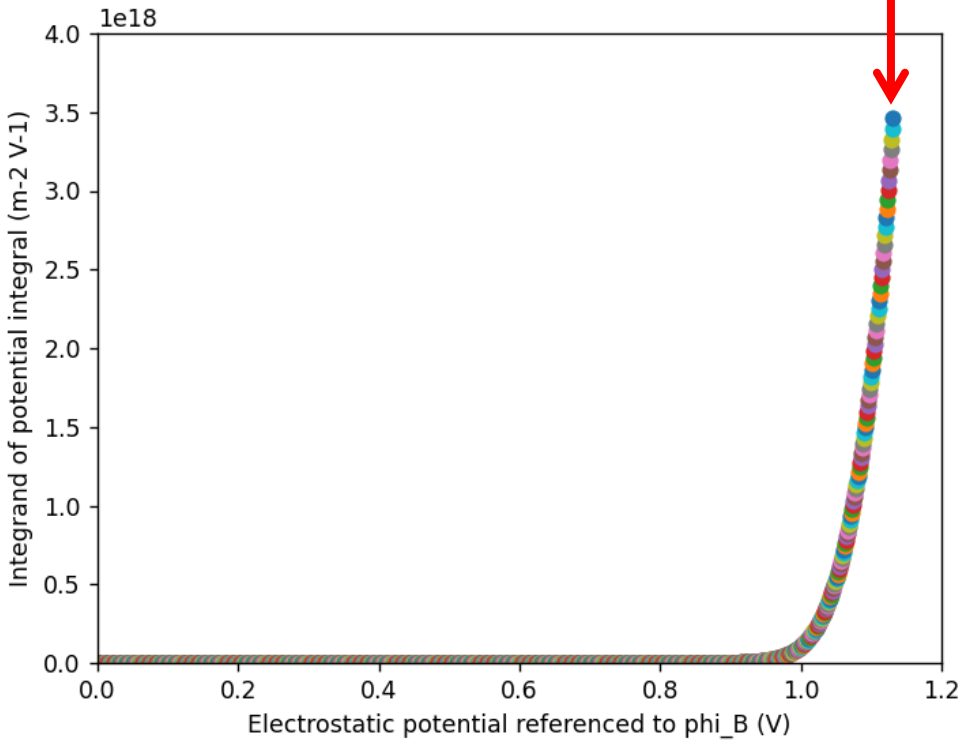
$$\approx \sum_{k=1}^{k_s} \frac{\left(n_i^2/N_a\right)\exp\left(\frac{q}{k_B T}\left(k\ \mathrm{mV}\right)\right)}{E_x\left(V_j + k\ \mathrm{mV}, V_j\right)} \times 0.1\mathrm{mV}$$

  - Here, $V_j + k_s$ mV is a good approximation of ϕ_s.
  - Parameters: V_{fb} is -1.08 V. C_{ox} is 2.88X10^{-6} F/cm^2. (t_{ox} is 1.2 nm.) N_a is 10^{18} cm^{-3}.

Write a code. (4)

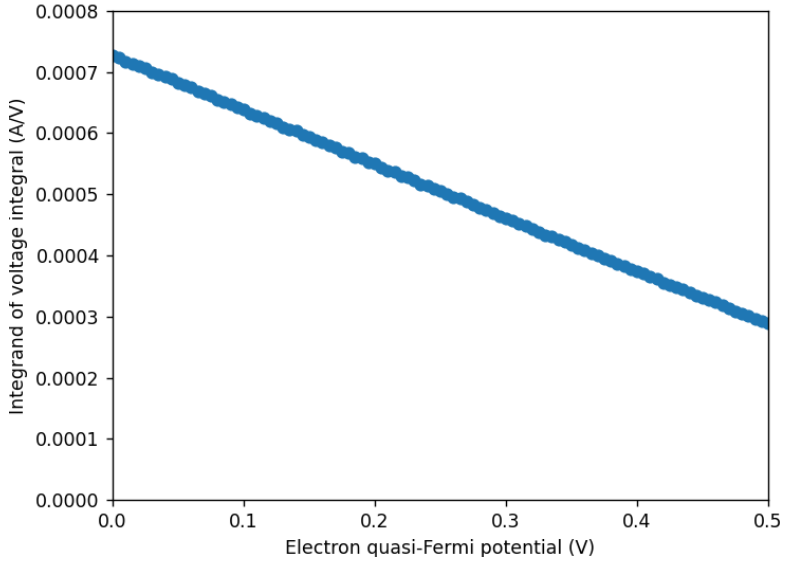
- A specific example, when V_j is 0 V.
  - Draw $\dfrac{(n_i^2/N_a)\exp\left(\frac{q}{k_B T}(\phi - V_j)\right)}{E_x(\phi, V_j)}$ as a function of ϕ.
  - Stopped when the MOS equation changes its sign. (Approximate ϕ_s)
  - Integrated electron density is about 1.517 X 10^{13} cm^{-2}.

Approximate ϕ

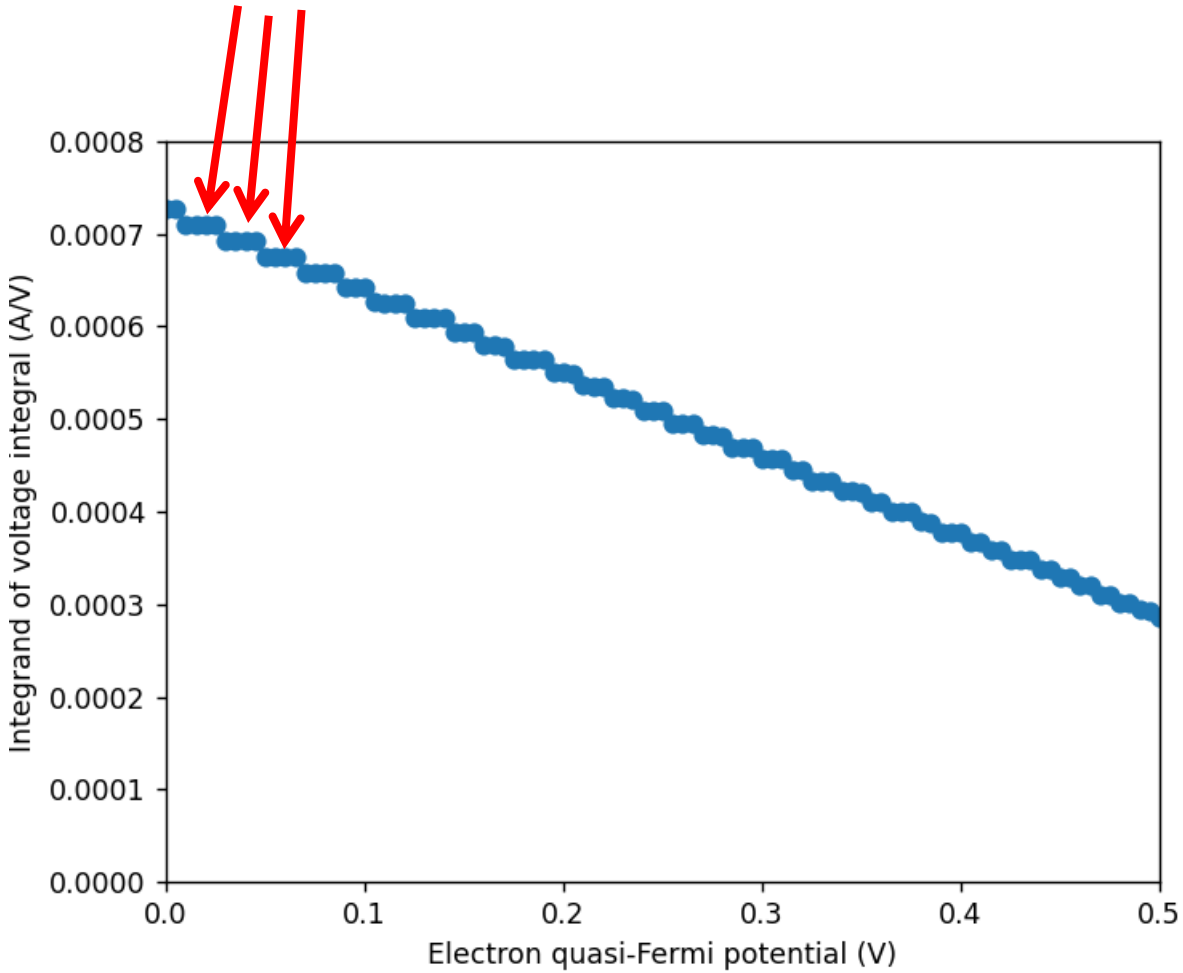


Write a code. (5)

- Integrand of the V-integral
  - V_{gs} = 1.1 V and V_{ds} = 0.5 V
  - By integration, we can find I_d = 252.8 μA. (1 mV ϕ spacing)
  - With 0.2 mV, I_d = 252.9 μA.



Step-wise???



IV curves

- Output curves
  - V_{gs} from 0.4 V to 1.1 V

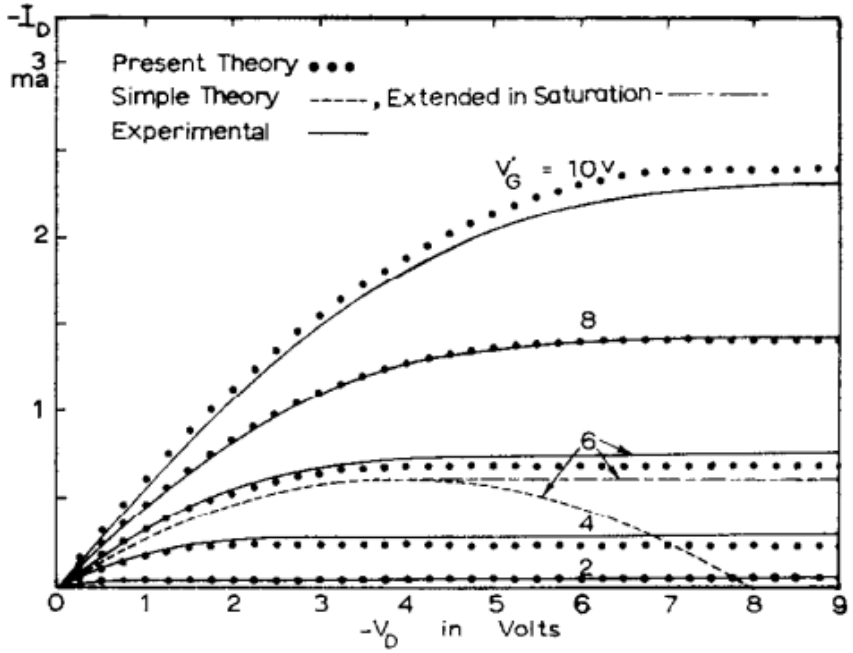
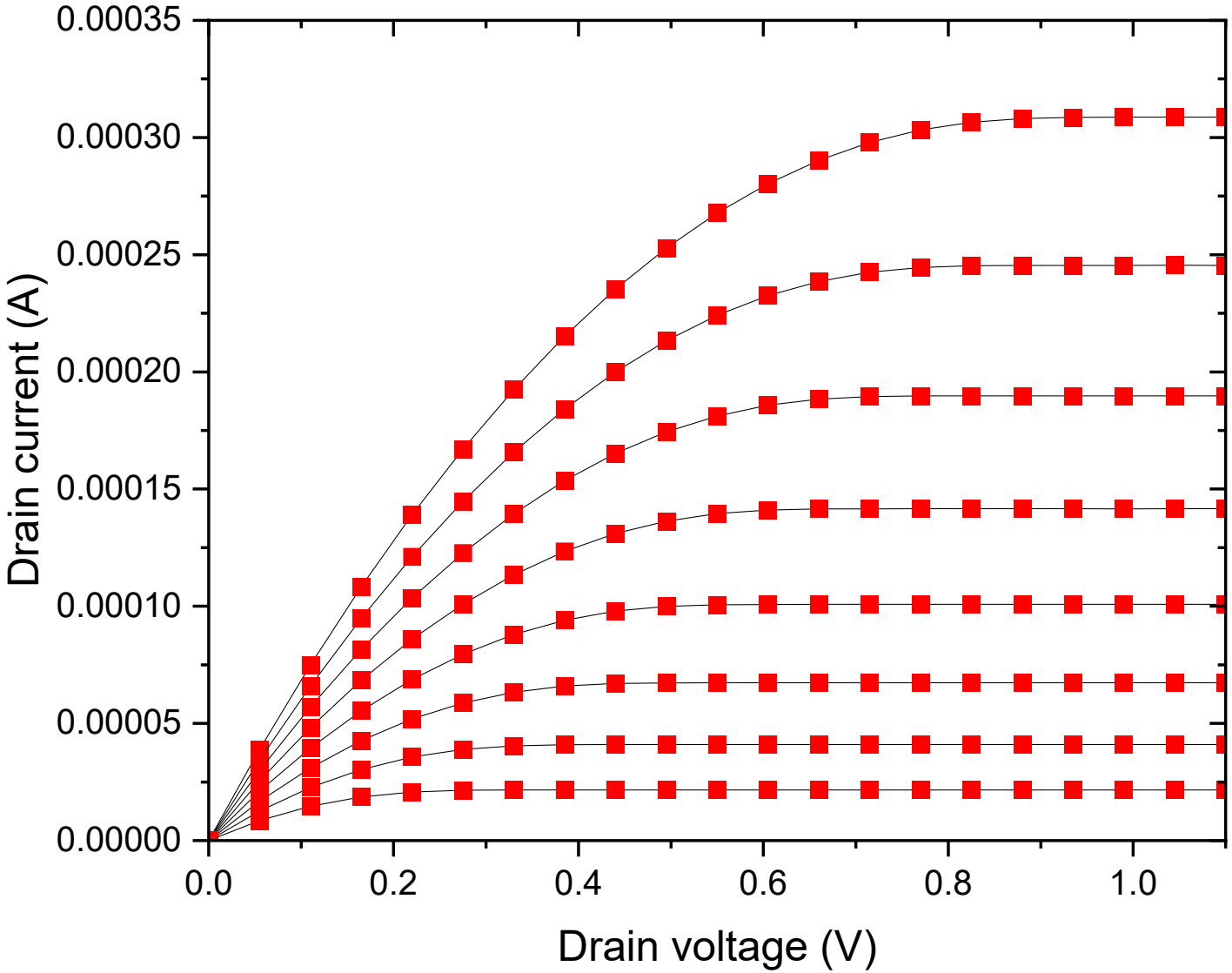


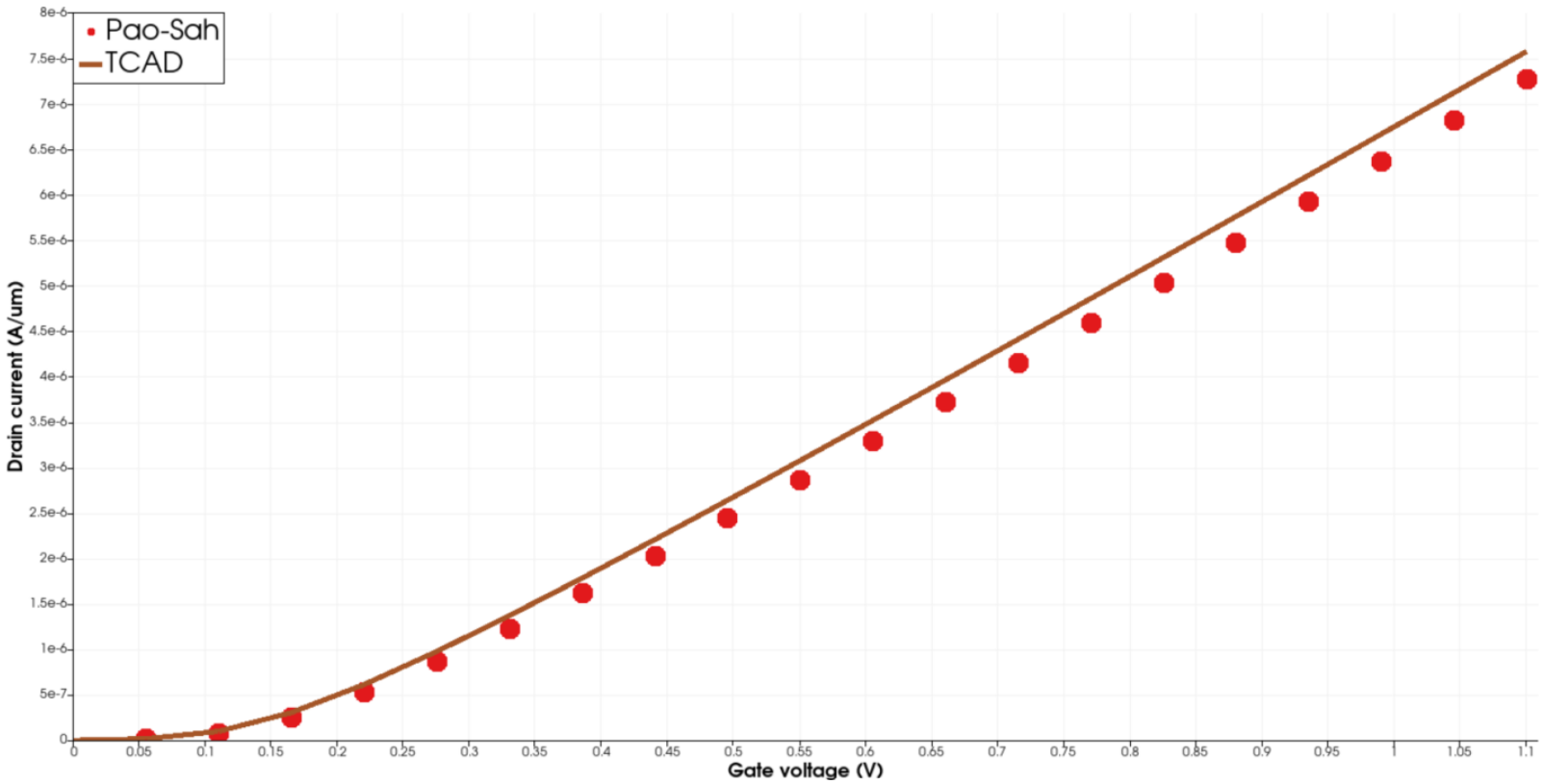
FIG. 2. The theoretical and experimental drain characteristics of a P-channel MOS transistor having $x_0 = 0{\cdot}2\ \mu$; $N_D = 4{\cdot}6 \times 10^{14}\ \mathrm{cm}^{-3}$; $\mu_p = 256\ \mathrm{cm}^2/\mathrm{V\text{-}sec}$ and area $= 8{\cdot}4 \times 10^{-4}\ \mathrm{cm}^2$.

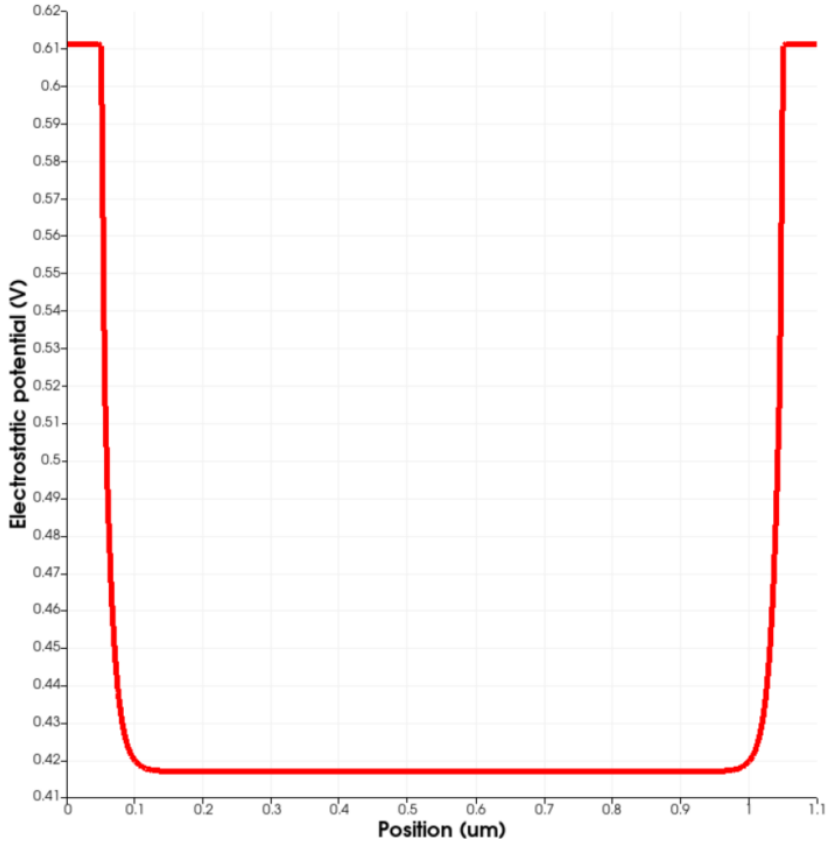
(H. C. Pao and C. T. Sah, SSE, 1966)



Verify the result.

- TCAD simulation result (at V_{ds} = 0.01 V)
  - They are similar, but not perfectly matched. *Why?*





Charge-sheet model

- Simpler model with further approximations
  - Consider the previous method to calculate Q_i:

$$Q_i = -q \int_{\delta}^{\phi_s} \frac{\left(n_i^2/N_a\right) \exp\left(\frac{q}{k_B T}(\phi - V)\right)}{E_x(\phi, V)} d\phi$$

  - A simpler way? Instead, Q_d is approximated as

$$Q_d = -qN_a W_d = -\sqrt{2\epsilon_{si} q N_a \phi_s}$$

Taur, Eq. (5.15)

  - Then, Q_i can be approximated as

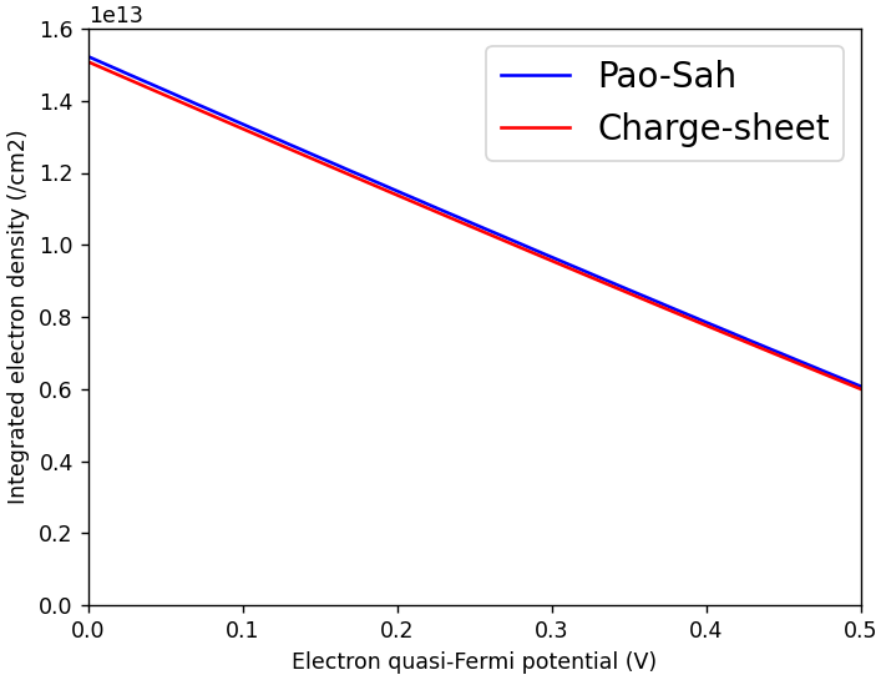
$$Q_i = Q_s - Q_d = -C_{ox}\left(V_{gs} - V_{fb} - \phi_s\right) + \sqrt{2\epsilon_{si} q N_a \phi_s}$$

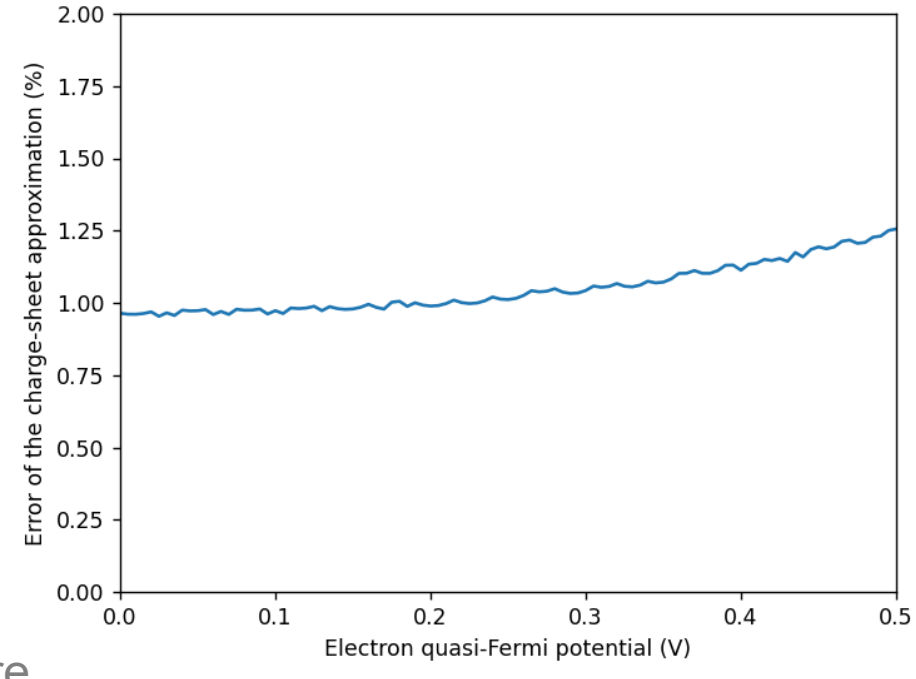
Taur, Eq. (5.16)

(Of course, it is not exact.)

Pao-Sah versus charge-sheet

- Consider a bias condition of V_{gs} = 1.1 V and V_{ds} = 0.5 V.
  - Calculate ϕ_s within a maximum error of 0.01 mV. Then, apply the charge-sheet approximation.
  - Only about 1 % error





Change of variable

- Now, Q_i can be a function of ϕ_s.
  - Variable change from V to ϕ_s:

$$I_d = \mu_{eff}\frac{W}{L}\int_0^{V_{ds}}[-Q_i(V)]dV = \mu_{eff}\frac{W}{L}\int_{\phi_{s,s}}^{\phi_{s,d}}[-Q_i(\phi_s)]\frac{dV}{d\phi_s}d\phi_s$$

Taur, Eq. (5.17)

Surface potentials at the two ends, $y = 0$ and L. They can be calculated by solving Taur, Eq. (5.14).

$V(\phi_s)$?

- Recall that

$$V_{gs} = V_{fb} + \phi_s + \frac{\sqrt{2\epsilon_{si} k_B T N_a}}{C_{ox}} \left[\frac{q\phi_s}{k_B T} + \frac{n_i^2}{N_a^2} \exp\left(\frac{q}{k_B T} (\phi_s - V) \right) \right]^{1/2}$$

Taur, Eq. (5.14)

  - We can rearrange

$$\frac{C_{ox}^2 \left(V_{gs} - V_{fb} - \phi_s \right)^2}{2\epsilon_{si} k_B T N_a} = \frac{q\phi_s}{k_B T} + \frac{n_i^2}{N_a^2} \exp\left(\frac{q}{k_B T} (\phi_s - V) \right)$$

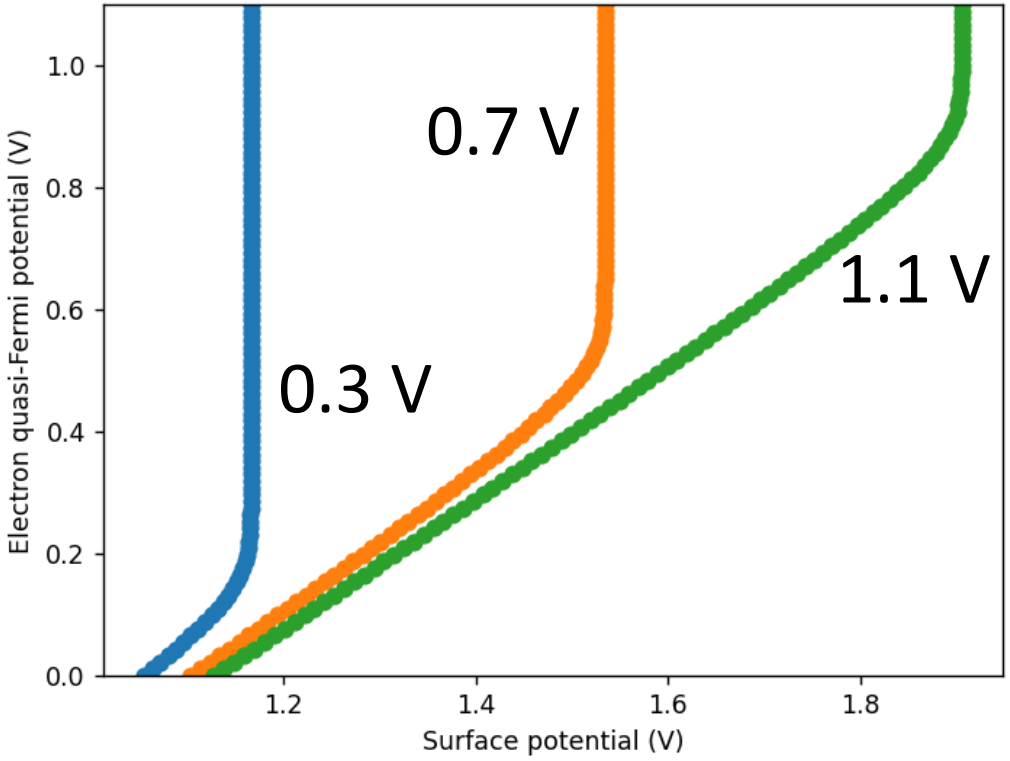
  - Therefore,

$$V = \phi_s - \frac{k_B T}{q} \log \left\{ \frac{N_a^2}{n_i^2} \left[\frac{C_{ox}^2 \left(V_{gs} - V_{fb} - \phi_s \right)^2}{2\epsilon_{si} k_B T N_a} - \frac{q\phi_s}{k_B T} \right] \right\}$$

Taur, Eq. (5.18)

Pao-Sah result

- Consider three gate voltages, V_{gs} = 0.3 V, 0.7 V, and 1.1 V.
  - Draw the electron quasi-Fermi potential (V) as a function of the surface potential (ϕ_s). See Taur, Fig. 5.3.



$\frac{dV}{d\phi_s}$? (1)

- Recall that

$$V = \phi_s - \frac{k_B T}{q} \log\left\{\frac{N_a^2}{n_i^2}\left[\frac{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right)^2}{2\epsilon_{si} k_B T N_a} - \frac{q\phi_s}{k_B T}\right]\right\}$$

Taur, Eq. (5.18)

  - Therefore,

$$\frac{dV}{d\phi_s} = 1 - \frac{k_B T}{q} \frac{-\frac{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right)}{\epsilon_{si} k_B T N_a} - \frac{q}{k_B T}}{\frac{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right)^2}{2\epsilon_{si} k_B T N_a} - \frac{q\phi_s}{k_B T}}$$

$\frac{dV}{d\phi_s}$? (2)

- Simple rearrange yields

$$\frac{dV}{d\phi_s} = 1 + \frac{2k_BT}{q}\frac{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right) + \epsilon_{si}qN_a}{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right)^2 - 2\epsilon_{si}qN_a\phi_s}$$

Taur, Eq. (5.19)

  - It is still very complicated…

Integrand

- When multiplied with $-Q_i(\phi_s)$,

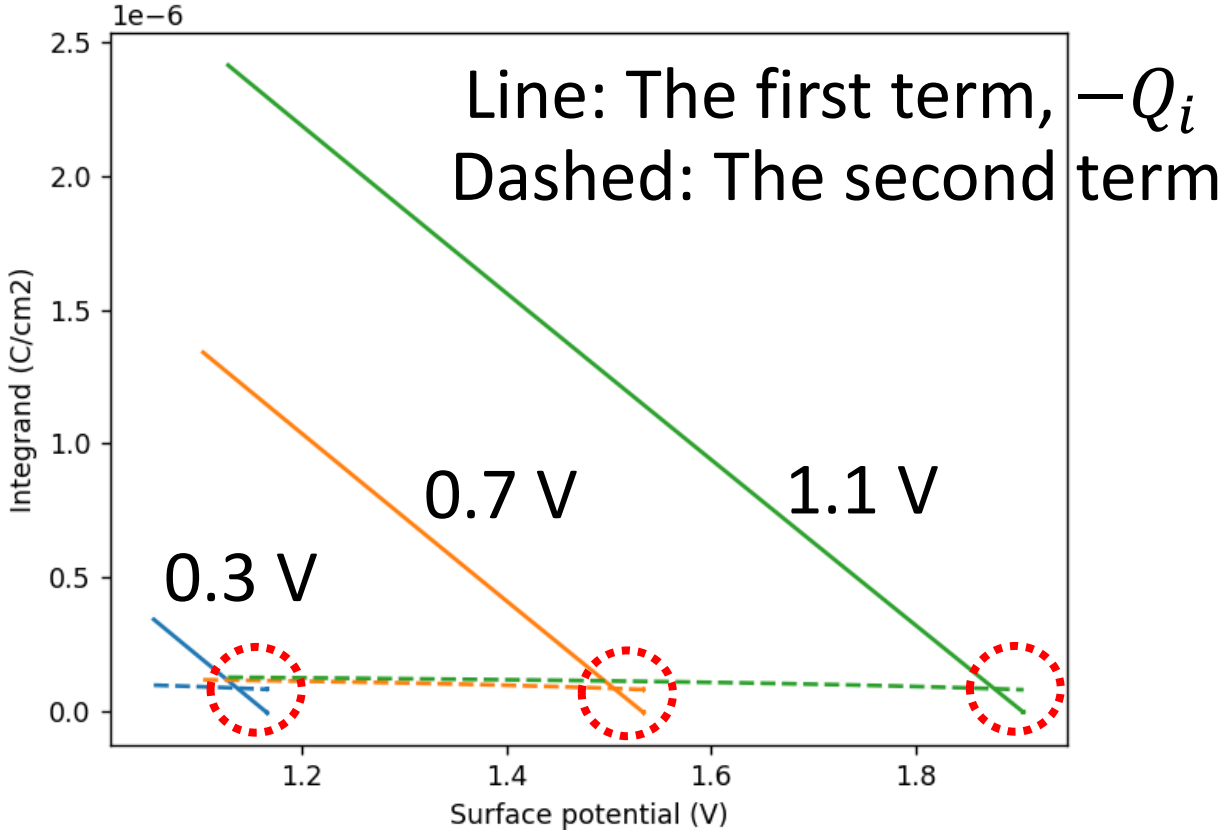
$$\left(-Q_i(\phi_s)\right)\frac{dV}{d\phi_s}$$

$$= -Q_i(\phi_s) + \frac{2k_BT}{q}\frac{C_{ox}^2\left(V_{gs} - V_{fb} - \phi_s\right) + \epsilon_{si}qN_a}{C_{ox}\left(V_{gs} - V_{fb} - \phi_s\right) + \sqrt{2\epsilon_{si}qN_a\phi_s}}$$

  - The second term is still very complicated…
  - *Is it really important?*

Comparison between two terms

- Let's draw two terms.
  - Assume that N_a = 10^{18} cm^{-3}, t_{ox} = 1.2 nm, and V_{fb} = -1.08 V.
  - The second term is small.
  - It is meaningful only when $Q_i \approx 0$.
  - This is corresponding to $C_{ox}\left(V_{gs} - V_{fb} - \phi_s\right) = \sqrt{2\epsilon_{si} q N_a \phi_s}$.



Integrand, again

- Within this condition,

$$\left(-Q_i(\phi_s)\right)\frac{dV}{d\phi_s} \approx -Q_i(\phi_s) + \frac{k_BT}{q}\frac{C_{ox}\sqrt{2\epsilon_{si}qN_a\phi_s} + \epsilon_{si}qN_a}{\sqrt{2\epsilon_{si}qN_a\phi_s}}$$

$$= -Q_i(\phi_s) + \frac{k_BT}{q}C_{ox} + \frac{k_BT}{q}\frac{\sqrt{2\epsilon_{si}qN_a}}{2\sqrt{\phi_s}}$$

Its integration yields $\frac{k_BT}{q}C_{ox}\phi_s$.

Its integration yields $\frac{k_BT}{q}\sqrt{2\epsilon_{si}qN_a\phi_s}$.

Drain current

- Using the previous approximation,
  - We can obtain the following expression:

$$I_d = \mu_{eff}\frac{W}{L}\left\{C_{ox}\left(V_{gs} - V_{fb} + \frac{k_B T}{q}\right)\phi_s - \frac{1}{2}C_{ox}\phi_s^2 - \frac{2}{3}\sqrt{2\epsilon_{si}qN_a}\phi_s^{1.5} + \frac{k_B T}{q}\sqrt{2\epsilon_{si}qN_a\phi_s}\right\}\Bigg|_{\phi_{s,s}}^{\phi_{s,d}}$$

Taur, Eq. (5.21)

  - Only with $\phi_{s,s}$ and $\phi_{s,d}$, we can evaluate the drain current.

Thank you!